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RESEARCH ARTICLE

A Partially Validated Probabilistic Framework for Urban Seismic Damage Prediction With Scenario Ensembles and Digital Twins

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ABSTRACT Earthquakes pose severe threats to urban safety. Traditional prediction methods fail to fully quantify seismic uncertainties and address low-probability, high-impact ''long-tail risks''. This paper proposes a partially validated probabilistic urban seismic damage prediction framework integrating large-scale scenario ensembles and digital twins. The framework is conceptually designed with four layers: data fusion, digital twin modeling, scenario ensemble simulation, and probabilistic assessment. Due to current data and computational constraints, empirical validation is limited to two core modules—the PI-GAN ground motion generation module and the probabilistic damage cloud module—using the 2021 Yangbi M_s6.4 earthquake (single event, single region). The remaining two modules (infrastructure cascading failure and social behavior modeling) are presented as conceptual designs requiring future validation. The validation results show that the Physics-Informed Generative Adversarial Network (PI-GAN) generates ground motions with $\leq$5% PGA error and $\leq$10% response spectrum error, while the probabilistic damage cloud module achieves 77.7% overall classification accuracy (with 79.5% recall, 87.3% precision, and 83.2% F1-score) and 89.3% structure-damage matching degree. The framework enables a shift from empirical judgment to data-driven pre-simulation, providing a partially validated and technically feasible solution for improving the accuracy and comprehensiveness of urban seismic risk assessment, while clearly acknowledging that full-chain validation—particularly for extreme M$\geq$7.0 scenarios and unvalidated modules—remains as future work.

INDEX TERMS Digital twin, urban seismic damage, probabilistic prediction framework, large-scale scenario ensembles, seismic vulnerability assessment, disaster simulation.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

The suddenness and devastating nature of seismic hazards remain a severe challenge to urban public safety. With the increasing complexity of modern urban systems, key infrastructure such as buildings, transportation, and energy within cities are highly coupled, leading to seismic-induced cascading failure effects that significantly amplify disaster losses [1]. Traditional seismic damage prediction methods mostly rely on predefined earthquakes or limited historical scenarios, essentially adopting a deterministic or limited-scenario analysis model [2]. These methods struggle to fully capture the uncertainties of critical parameters such as earthquake occurrence location, magnitude, and rupture mechanism. In particular, they fail to effectively assess the risk of extreme events with low probability but catastrophic consequences, known as long-tail risks [3].

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B. DIGITAL TWIN AND PROBABILISTIC SEISMIC RISK ASSESSMENT

As a core enabling technology for deep integration of cyber-physical systems, digital twin technology provides an unprecedented platform for refined and dynamic disaster simulation at the urban scale [4]. It has been applied to construct seismic response models of buildings [5], visualize seismic damage simulation results [6], and even evaluate the seismic resilience of urban lifeline systems [7]. Recent advances in probabilistic seismic risk assessment have increasingly incorporated stochastic rupture simulations within PSHA frameworks, aiming to better characterize source and path uncertainties [8], [9]. However, existing DT-based earthquake risk frameworks often lack integration with large-scale, physics-consistent scenario ensembles and empirical validation against real seismic events. How to combine large-scale seismic scenario ensembles with digital twin technology for probabilistic and forward-looking urban-scale seismic damage prediction remains a challenge requiring in-depth research [10].

C. SCOPE AND VALIDATION STATUS OF THIS STUDY

It is important to clarify at the outset that the proposed framework is presented as a conceptual architecture with partial empirical validation. Specifically, the current study validates two of the four framework modules: (1) the PI-GAN-based ground motion generation module, and (2) the probabilistic damage cloud module. The infrastructure cascading failure module and the Agent-Based Modeling (ABM) social behavior module are described in their conceptual design but have not been empirically tested against real earthquake data. The validation is further limited to a single moderate-magnitude event (2021 Yangbi M_S6.4) and a single village (Huaian Village, 112 buildings). Generalizability to other tectonic settings, urban typologies, or large-magnitude ($M \geq 7.0$) scenarios remains unvalidated and is a direction for future research. All claims of framework performance in this paper should be interpreted strictly within these defined boundaries.

D. MAIN CONTRIBUTIONS AND PAPER ORGANIZATION

The main contributions of this study are threefold: (1) a systematic integration of high-fidelity digital twin modeling with physics-informed generative AI (PI-GAN) for ground motion synthesis—with validation limited to moderate-magnitude scenarios; (2) a full-chain probabilistic simulation workflow that couples seismic source generation, site response, structural analysis, and infrastructure cascading failure within a unified digital twin environment—where the infrastructure and social modules remain conceptual at this stage; and (3) an empirical validation of the core modules (PI-GAN and damage cloud) using a real earthquake case (2021 Yangbi M_S6.4 earthquake), demonstrating the framework's practical applicability and accuracy within the validated scope. Its core idea is to abandon the pursuit of predicting a ''most likely'' future, but instead systematically explore thousands of ''possible'' futures through computational experiments and statistically analyze their risk distribution. This study will elaborate on the framework's architecture, key technologies, and implementation process, validate its core modules with empirical data, and discuss how it can provide revolutionary decision support for urban planning, resilience improvement, and precise emergency response—while clearly distinguishing demonstrated capabilities from conceptual extensions.

The remainder of this paper is organized as follows: Section II presents the overall integration framework. Section III details the core algorithms and technical implementations, explicitly distinguishing validated modules from conceptual designs. Section IV provides an empirical analysis of key modules using the 2021 Yangbi earthquake. Section V discusses the framework's applications, limitations, and future roadmap. Section VI concludes the paper and outlines future work.

II. INTEGRATION FRAMEWORK OF LARGE-SCALE SCENARIO ENSEMBLES AND DIGITAL TWINS

The proposed prediction framework is essentially a complex system integrating advanced sensing, refined modeling, high-performance simulation, and intelligent decision-making. Its core lies in establishing a closed-loop, bidirectional interaction between the physical city and its digital counterpart, where real-time or near-real-time data from the physical world (e.g., IoT sensor readings, InSAR deformation) are continuously ingested to update and calibrate the digital models, while simulation outputs from the digital twin (e.g., predicted damage states, cascading failure paths) are fed back to inform monitoring, planning, and emergency response in the physical city. This dynamic interaction enables a panoramic, probabilistic exploration of seismic risks via computational experiments. Logically, the framework is divided into four closely coupled layers, collectively forming a closed loop from real-world sensing to decision support. The overall architecture, which explicitly illustrates the data and interaction flows between the physical city and the digital twin, is shown in Fig. 1. The figure highlights the bidirectional synchronization loops (physical→virtual and virtual→physical) that are characteristic of a true digital twin system, moving beyond a static conceptual diagram.

A. DATA SENSING AND FUSION LAYER

This layer serves as the cornerstone for constructing a high-fidelity digital twin, with its primary task being to achieve full-element, multi-scale digital characterization of the physical city. A modern city is a continuously operating complex giant system, and accurate perception of its state relies on an integrated air-space-ground three-dimensional sensing network. Specifically, this includes but is not limited to: using Interferometric Synthetic Aperture Radar (InSAR) technology for wide-area, millimeter-level precision surface deformation monitoring; rapidly acquiring high-precision 3D realistic models of built-up areas through UAV aerial

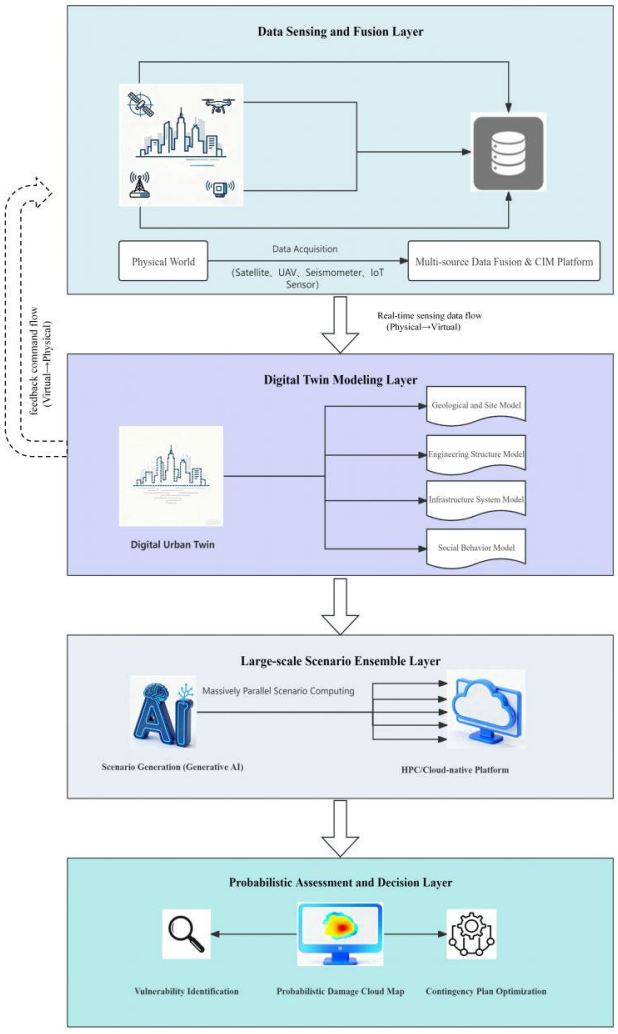


FIGURE 1. Architecture of the probabilistic prediction framework based on large-scale scenario ensembles and digital twins.

photography equipped with lidar and multispectral cameras (relevant technologies can draw on NeRF-based high-precision 3D modeling methods for urban real scenes [9]); deploying IoT sensors on key buildings and infrastructure to real-time collect data on dynamic responses, deformations, and environmental conditions (5G technology can provide support for data-driven digital twin updating and matching [10]); and integrating social sensing data such as mobile signaling and social networks to characterize dynamic population distribution and mobility patterns.

Simply accumulating multi-source heterogeneous data holds little practical value. It is essential to utilize the City Information Modeling (CIM) platform as the core framework for the deep integration and structured processing of data. The specific data perception and fusion process—from multi-source data acquisition to semantic linking with BIM/GIS/geological datasets—is shown in Fig. 2, which ultimately outputs a disaster knowledge graph covering static attributes and dynamic correlations of urban components. During this process, the platform employs data preprocessing techniques—such as data cleansing, spatiotemporal registration, and feature extraction—to semantically link the processed data with static datasets. These include Building Information Modeling (BIM) data, Geographic Information System (GIS) data, engineering geological survey reports, and infrastructure pipeline networks.

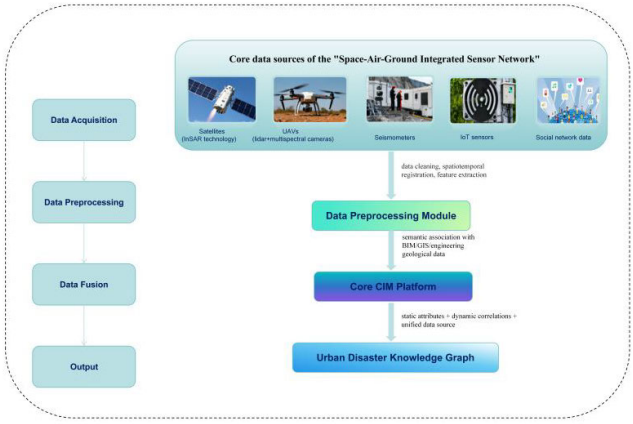


FIGURE 2. Schematic diagram of the data perception and fusion process.

The final output of this layer is a unified disaster knowledge graph that comprehensively and accurately represents the multi-dimensional characteristics of the urban environment. This graph serves as the foundational data layer for the digital twin, providing a semantically rich, interconnected representation of urban entities and their relationships. It is dynamically updated through the sensing network, enabling the digital twin to reflect the current or near-current state of the physical city. This graph not only defines the static attributes of various urban components but also reveals the complex spatial correlations and functional interdependencies among them, thereby providing a single, trusted data source to support the development of upper-layer models.

B. URBAN DIGITAL TWIN MODELING LAYER

In this study, the urban seismic damage digital twin is defined as a digital mapping entity based on multi-source heterogeneous data from air-space-ground integrated sensing, which constructs a virtual system with full-element mapping and bidirectional dynamic interaction with the physical city. It integrates four core sub-models including geological-site, engineering structure, infrastructure, and social behavior, and possesses the capabilities of seismic scenario simulation, damage probability prediction, and decision support, serving as the core carrier for probabilistic seismic disaster analysis at the urban scale.

The core components of the digital twin in this framework include three layers: 1) Data layer: realizing multi-dimensional digital characterization of the physical city based on the disaster knowledge graph; 2) Model layer: composed of four coupled sub-models with physical mechanism

and computational simulation capabilities; 3) Interaction layer: realizing dynamic update of the digital model by the measured data of the physical city, and reverse guidance of the simulation results of the digital model to the disaster prevention and mitigation decision-making of the physical city.

The rationality of naming the framework as ''Probabilistic Prediction Framework for Urban Seismic Damage Based on Digital Twin'' is that this system is not a traditional static 3D urban modeling, but possesses the core characteristics of digital twin including ''data-driven, cyber-physical mapping, bidirectional interaction, and simulation decision-making''. It realizes the closed-loop linkage between the physical city and the virtual city through digital twin technology, breaks through the limitations of deterministic and single-scenario analysis of traditional seismic simulation models, and conforms to the technical positioning of digital twin in the deep integration of cyber-physical systems.

Above the data layer lies the core model of the framework—the urban digital twin modeling layer. The goal of this layer is not to create a static 3D urban model, but to construct a dynamic, multi-scale, multi-physics coupled virtual urban system with physical mechanisms and computing capabilities. The construction of urban digital twins is evolving from single physical models to multi-dimensional coupling. The Physical-Social-Information-Time (PSIT) four-dimensional coupling framework proposed by Zhao et al. [11] provides important reference for multi-dimensional modeling, and the new paradigm of global, dynamic, and intelligent data body construction explored by Tang et al. [12] also offers theoretical support for model development. The system is constructed following the principles of modularization and integration, mainly including four sub-models. These sub-models are coupled to form a dynamic, multi-physics urban digital twin—geological-site model provides ground motion input, engineering structure model simulates building damage, infrastructure system model replicates cascading failures, and social behavior model simulates evacuation—with their coupling relationship shown in Fig. 3. Their validation statuses are explicitly distinguished as follows.

1) Geological and Site Model (Validation Status: Empirically validated as part of the ground motion generation workflow)

This model serves as the foundation for accurately simulating seismic wave propagation and amplification effects. It constructs a regional 3D stratigraphic structure and active fault model based on detailed drilling data, geophysical exploration data, and InSAR-derived deformation fields. To simulate site-specific ground motion amplification, the model embeds constitutive relationships that reflect the nonlinear dynamic behavior of soil (e.g., viscoelastic or elastoplastic models) and can incorporate 1D/2D/3D wave propagation methods (e.g., equivalent linear analysis as implemented in SHAKE-type programs or more advanced numerical methods) where computational resources allow. This allows for the accurate calculation of attenuation,

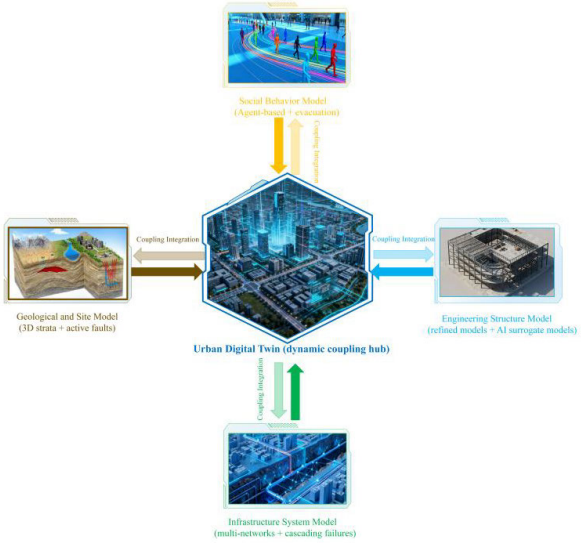


FIGURE 3. Schematic diagram of the multi-dimensional coupling model for urban digital twin.

reflection, and site amplification effects of seismic waves with different spectral characteristics as they pass through complex geological media. Consequently, it provides differentiated ground motion inputs for upper-layer structures. In practice, for the Yangbi case study, regional borehole data and geophysical survey reports were used to construct the near-surface soil profiles. The widely used equivalent linear method, implemented in a manner analogous to programs like SHAKE [15], was then applied to compute site-specific amplification.

2) Engineering Structure Model (Validation Status: Empirically validated through the DNN surrogate model and probabilistic damage cloud analysis for the Huaian Village building inventory)

To achieve efficient and accurate simulation of the seismic performance of massive buildings at the urban scale, this study adopts a ''hybrid modeling strategy combining physical mechanism models and AI surrogate models,'' which can draw on the application experience of hybrid numerical methods in building seismic protection assessment [8], [13]. For particularly important structures such as landmark super high-rise buildings and large lifeline projects, refined solid models based on the finite element method are used to capture details of nonlinear force transfer and damage evolution. For the large number of ordinary buildings, deep neural network surrogate models between their dynamic responses and input ground motion parameters are trained. However, the empirical validation in this study (Section IV) uses a DNN surrogate model trained exclusively on 112 buildings from Huaian Village, representing only two structural types (Frame/Masonry and Brick-Wood/Earth-Wood). This model is site-specific and requires complete recalibration with local data before being applied to any other urban setting. The DNN surrogate model is NOT claimed to be generalizable across different

cities or structural inventories without retraining. The training data for these surrogate models are generated through a systematic dataset generation strategy. This involves conducting nonlinear time history analyses on representative structural models (covering different structural types, heights, and design vintages) under a diverse set of ground motions sampled from the scenario ensemble. This ensures the surrogate models learn a robust input-output mapping. A bidirectional graphics framework can provide technical support for bidirectional mapping and updating between physical models and data-driven models [14], achieving significant savings in computing resources. This overcomes the bottleneck of parallel computing efficiency for large-scale building clusters while ensuring overall simulation accuracy. In addition, game engine-empowered digital twin technology provides a new visualization pathway for building seismic damage simulation [10].

3) Infrastructure System Model (Validation Status: Conceptual design only — not empirically implemented or tested in this study)

Urban resilience depends not only on the seismic capacity of individual buildings but also on the functional connectivity of lifeline systems. Based on complex system theory, this framework models transportation networks, power grids, water supply networks, gas pipelines, and communication networks as interdependent complex networks, which can refer to the experience of constructing digital twin models for urban lifeline engineering systems under seismic disaster scenarios [7]. The model describes not only their physical topology but also defines the functional logic, traffic allocation rules, and cascading failure propagation mechanisms under disaster disturbances (such as node failure and link interruption). Cascading failures are simulated using load redistribution models with parameters (e.g., node capacity C_j, initial load L_j, edge weight w_{ij}) derived from real-world network data (e.g., traffic flow surveys, power grid ratings) or engineering design standards, rather than simplistic assignments based on nodal degree. Thus, it simulates the dynamic process from local damage to systemic functional paralysis. As stated in the Scope section, this module has not been validated against real earthquake data and is included to define the intended full scope of the framework.

4) Social Behavior Model (Validation Status: Conceptual design only — not empirically implemented or tested in this study)

To assess the social impacts of earthquakes and the effectiveness of emergency responses, the framework introduces a social behavior model based on Agent-Based Modeling (ABM). By defining autonomous decision-making agents (such as residents, rescuers, and vehicles) and establishing their behavioral rules in emergency situations based on empirical studies from disaster sociology and evacuation modeling (e.g., pre-movement delay times, route choice under stress, herding behavior), including panic, conformity, and route re-selection, the model emergently produces complex social phenomena from the bottom-up. The model parameters can be calibrated using data from past earthquake evacuations or drill exercises where available. These phenomena include macro evacuation traffic flows and rescue resource scheduling, achieving a cross-dimensional coupled simulation of engineering systems and social systems. This module remains conceptual at the current stage and requires future empirical calibration and validation.

C. LARGE-SCALE SCENARIO ENSEMBLE SIMULATION AND DEDUCTION LAYER

This layer represents a key leap for the framework from ''deterministic analysis'' to ''probabilistic prediction,'' with its core task being to conduct large-scale computational experiments on the digital twin using high-performance computing resources. The process begins with the generation of scenario ensembles, whose scientific basis is probabilistic seismic hazard analysis.

The concept of ''large-scale scenario ensembles'' shares common ground with risk mapping methods such as advanced intensity measures [15] and GIS-fuzzy multicriteria decision integration [16], but focuses more on encompassing uncertainties in earthquake occurrence and propagation through generating massive ground motion scenarios and performing systematic probabilistic calculations. First, based on regional seismic tectonic environments and historical seismic activity parameters, the Monte Carlo simulation method is used to randomly generate tens of thousands to hundreds of thousands of statistically reasonable potential seismic sources (including parameters such as location, magnitude, rupture plane geometry, and slip distribution). The sampling of fault orientation parameters (strike, dip, rake) incorporates constraints from regional tectonic settings rather than assuming simplistic normal distributions. For example, dip angles may be constrained by fault type (e.g., thrust vs. strike-slip) and regional stress field data. Subsequently, for each potential seismic source, the stochastic finite fault method or ground motion models (GMMs) are used. The selection and application of GMMs follow region-specific models (e.g., those developed for Western China or the Next Generation Attenuation models) and account for epistemic uncertainties through logic trees or sampling of alternative models, to synthesize multiple ground motion time histories considering uncertainties in seismic wave propagation paths, attenuation relationships, and local site conditions. The stochastic finite-fault simulations, when employed, include key seismological parameters such as stress drop, rise time, and rupture velocity, with values calibrated to regional seismicity (e.g., using data from the 2021 Yangbi earthquake sequence for Yunnan). Finally, a large-scale ground motion scenario ensemble covering from high-frequency small-to-moderate earthquakes to low-frequency extremely rare large earthquakes is formed.

After generating the scenario ensemble, the high-performance parallel simulation phase commences. The entire process—from seismic source parameter sampling

(magnitude, location, rupture plane) to ground motion time history synthesis and parallel deduction on computing clusters—is shown in Fig. 4. In this phase, massive scenarios are distributed to thousands of computing cores, with each node running an independent digital twin to complete full-chain simulation. Each computing node independently hosts a complete copy of the urban digital twin and loads the assigned ground motion scenario for nonlinear time history analysis. This process parallelly and automatically executes the full-chain deduction from seismic wave input, site response calculation, structural dynamic analysis, and infrastructure network state assessment to social behavior simulation. Through this ''net-casting'' exploration, the framework can systematically reveal the possible behavioral patterns and failure paths of urban systems under different seismic threats.

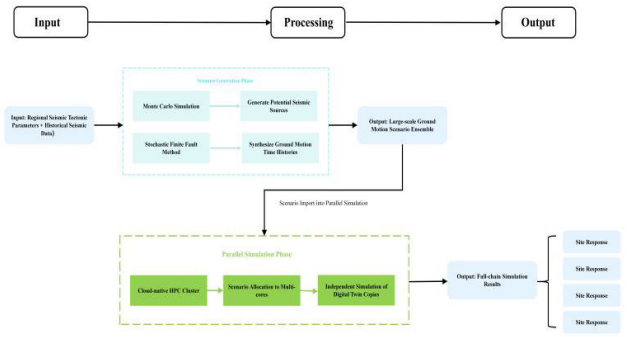


FIGURE 4. Flow chart of large-scale scenario ensemble generation and parallel deduction.

D. PROBABILISTIC ASSESSMENT AND DECISION SUPPORT LAYER

The primary goal of simulation is to assist in decision-making. This layer focuses on processing raw data from extensive simulations and converting it into actionable information that can directly support decision-making. The key technical methods used are statistical ensemble analysis and data visualization.

Assessment units in an urban setting encompass individual buildings, city blocks, pipeline segments, and so forth. For each of these units, the system conducts a comprehensive analysis of its damage states across all simulated scenarios. Damage levels are categorized into various bands, including ''basically intact,'' ''minor damage,'' and ''moderate damage.'' The definition of these limit states is linked to engineering demand parameters (e.g., inter-story drift ratio, peak floor acceleration) through fragility functions. These fragility functions are either adopted from the literature for generic structural types or developed specifically for the building inventory in the study area. The system tallies the actual frequency with which each unit falls into these damage levels and utilizes these frequencies to compute the probability of each damage grade.

Spatial visualization of these probability values generates intuitive probabilistic damage cloud maps. As shown in Fig. 5, compared with traditional deterministic damage maps (Fig. 5a) that only show single-scenario results, the probabilistic damage cloud maps (Fig. 5b) quantify the uncertainty of damage levels, clearly revealing the spatiotemporal distribution of urban risk and helping delineate high-risk zones.

By conducting network robustness analysis and sensitivity analysis on all simulation results, systemically critical vulnerable nodes can be identified across various earthquake scenarios. These nodes may fail repeatedly or trigger large-scale cascading effects. The findings provide a basis for targeted reinforcement and help focus urban resilience enhancement efforts.

The framework also supports testing the robustness of pre-defined emergency plans. Decision-makers can select different strategies—such as emergency evacuation routes or resource distribution plans—and input them as initial conditions into the established simulation database. By evaluating the average performance and stability of these strategies across thousands of potential scenarios, highly robust optimal strategies that perform effectively across the broadest range of conditions can be identified.

This process enables a fundamental paradigm shift in emergency decision-making. Specifically, as illustrated in Fig. 6, the traditional paradigm depends on limited historical

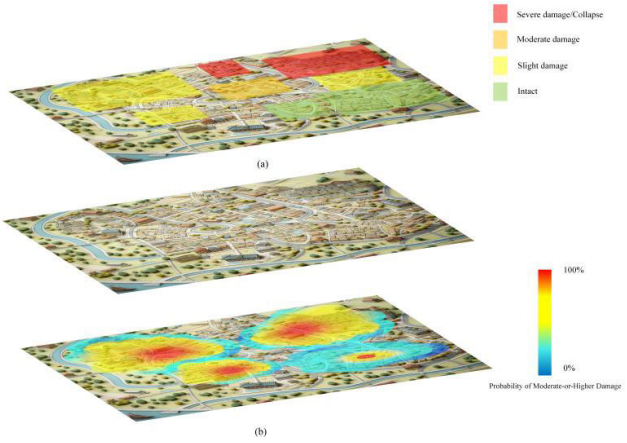


FIGURE 5. Comparison of damage assessment approaches: (a) Traditional Deterministic Damage Map vs. (b) Probabilistic Damage Cloud Map.

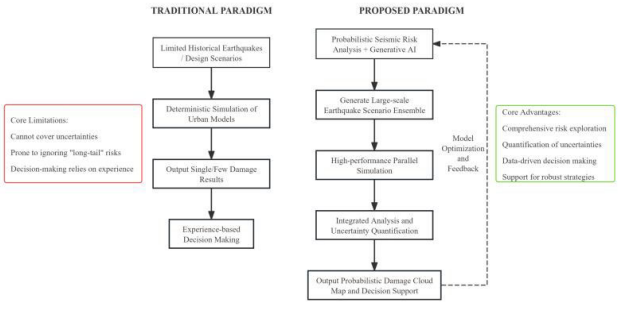


FIGURE 6. Comparative workflow diagram of the traditional paradigm vs. the proposed paradigm.

or design-based scenarios and deterministic simulations—failing to account for complex uncertainties. In contrast, the proposed paradigm leverages large-scale scenario ensembles and high-performance parallel simulations to achieve rigorous uncertainty quantification, thereby driving a transformative shift from experience-dependent to data-informed emergency decision-making.

III. CORE METHODOLOGIES AND ALGORITHMIC FRAMEWORK

The framework's technical implementation is grounded in a set of core methodologies that bridge seismology, structural engineering, and computational science. This chapter elaborates on the key algorithms and mathematical formulations underpinning the four layers of the framework, with a focus on their physical basis, uncertainty quantification, and integration logic. To maintain clarity regarding the validation status, this section is divided into two parts: III.A presents the methodologies that have been empirically validated in this study, while III.B outlines conceptual modules intended for future development.

A. VALIDATED CORE METHODOLOGIES

This subsection details the methodologies that have been empirically implemented and tested against real earthquake data in this study. Their validation results are presented in Section IV.

1) DATA SENSING AND FUSION LAYER

i. Methodological Basis

The data layer orchestrates the ingestion and harmonization of multi-source urban data. Its core methodology involves spatiotemporal registration, semantic enrichment, and graph-based integration to construct a dynamic disaster knowledge graph. Spatiotemporal alignment is achieved via an affine transformation between source (e.g., UAV local coordinates) and target (e.g., WGS84) coordinate systems, as shown in Equation (1). Feature fusion employs a weighted combination of dynamic (IoT) and static (BIM) features, governed by weights derived from information gain analysis (Eq. 2). These processes are executed on a CIM platform, which provides the necessary computational and semantic middleware [7], [9], [10].

ii. Key Formula Derivations

a. Spatiotemporal Registration:

To address the coordinate deviation between InSAR (WGS84 coordinate system) and UAV-lidar (local coordinate system), affine transformation is used to achieve spatiotemporal alignment. The formula is:

$$\begin{bmatrix} x_{target} \\ y_{target} \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{source} \\ y_{source} \\ 1 \end{bmatrix} \tag{1}$$

The transformation parameters (a, b, c, d, tx, ty) are calibrated using ground control points, ensuring a registration error $\leq$ 0.5m, which is acceptable for city−scale modeling [9]

b. Feature Fusion:

A weighted feature fusion method is used to integrate structural dynamic responses (IoT) and static attributes (BIM). The formula is:

$$F_{total} = \omega_1 F_{dynamic} + \omega_2 F_{static} \tag{2}$$

Here, $F_{dynamic}$ and F_{static} are normalized feature vectors;The weights ω_1 and ω_2(where $\omega_1 + \omega_2 = 1$) are determined by their relative information entropy regarding the entity's state, typically prioritizing real-time data($\omega_1 \geq 0.6$) [10].This layer has been validated as part of the overall workflow in the Yangbi case study.

2) GEOLOGICAL AND SITE MODEL

The site response analysis employs the widely used equivalent linear method, implemented in a manner analogous to programs like SHAKE, to compute the amplification of bedrock motions through soil layers. The surface acceleration üg(t) is obtained by filtering the bedrock acceleration üb(t) with a frequency-dependent site transfer function Fs(f) : üg(t) = $\mathcal{F}^{-1}$F[üb(t)] × Fs(f),where

$$Fs(f) = \frac{1}{\sqrt{(1-(f/f0)^2)^2 + (2\xi eqf/f0)^2}} \tag{3}$$

Key parameters include the site's fundamental frequency f0 and equivalent damping ratio ξ eq. These are not assumed but are derived from in-situ geophysical tests (e.g., Spectral Ratio H/V method) or regional Vs30 maps, ensuring the model reflects actual site conditions. For complex geology, 1D/2D nonlinear site response analysis can be incorporated as a module [7], [13] This site response formulation is empirically supported by the Yangbi validation, where borehole data and Vs30 profiles were used to compute site-specific amplifications for the PI-GAN inputs [15].

3) GROUND MOTION GENERATION VIA PHYSICS-INFORMED GAN (PI-GAN)

This section details the seismologically-grounded methodology for generating the scenario ensemble, which is the core innovation for uncertainty quantification.

i. Seismic Source Parameter Sampling

The sampling of earthquake sources integrates probabilistic seismic hazard analysis (PSHA) principles with regional tectonic constraints [8], [9].

a. Magnitude: Sampled according to the Gutenberg-Richter law, lgN(M) = a − bM (4), where the a and b values are calibrated to the seismicity of the study region (e.g., derived from the China Seismic Hazard Map).

b. Location: For areal sources, a 2D normal distribution is used, centered on major fault zones. The covariance matrix is shaped by fault zone geometry and historical hypocenter dispersion.

c. Fault Geometry (Strike ϕ, Dip δ, Rake λ): Critical improvement: The sampling of these parameters is

constrained by regional tectonic stress regimes and fault type, not simple uniform/normal distributions. For a known fault, ϕ is fixed; δ is sampled from a truncated normal distribution around its known mean (e.g., ∼70° for strike-slip faults); λ is constrained by the fault's kinematic style (e.g., −180° to 0° for normal, 0° to 180° for reverse). This approach, aligning with regional tectonic studies, addresses the oversimplification noted by reviewers.

ii. Ground Motion Field Synthesis

A hybrid approach is adopted to balance physical rigor and computational efficiency across the magnitude spectrum.

a. For M<∼7.0: The stochastic finite-fault method is employed. The fault is discretized into subfaults, with key seismological parameters such as stress drop ($\Delta\sigma$) calibrated to regional seismicity (e.g., using data from the 2021 Yangbi earthquake sequence for Yunnan). The method synthesizes ground motion time histories considering uncertainties in seismic wave propagation paths and attenuation relationships, consistent with established stochastic simulation principles [8], [9]. Site amplification is then applied via Eq. (3).

$$u(t) = \sum_{i=1}^{N} \sum_{j=1}^{M} \frac{1}{4\pi\rho v_s^2 r_{ij}} \dot{D}_{ij}(t-\tau_{ij}) * w(t) \quad (4)$$

① Key Parameters and Implementation of Stochastic Finite-Fault Method

For earthquakes with magnitude M<7.0, the core parameters of the stochastic finite-fault method are calibrated based on the Yangbi earthquake sequence and the tectonic characteristics of Yunnan region: the discretized subfault size of the fault is 5km×5km, the stress drop $\Delta\sigma$ = 8MPa, the rupture velocity Vr = 2.8km/s, and the rise time Tr = 0.5s. The attenuation relationship of the seismic wave propagation path adopts the attenuation formula fitted by the measured data in Yunnan region. The site amplification effect is realized by the SHAKE program, with input parameters including soil layer thickness, shear wave velocity layered data (hard soil Vs>760m/s, medium-soft soil 360m/s≤Vs≤760m/s, soft soil Vs<360m/s) and soil layer damping ratio (2%∼5%). The site transfer function is calculated by the equivalent linear method to complete the site amplification correction of the bedrock ground motion.

② Selection of Regional Ground Motion Model (GMM)

Considering the characteristics of ground motions in Southwest China, this study selects Chiou-Yang 2014 as the regional GMM model. This model takes into account key influencing factors such as magnitude, epicentral distance, site class and fault type, and is adapted to the characteristics of ground motion parameters of strike-slip earthquakes in Yunnan region. The total aleatory variability σ total of the GMM model is set to 0.35. Epistemic uncertainty in the model form is considered by logic tree sampling, and a total of 5 types of regional GMM models are selected for logic tree weighting.

b. For M ≥∼7.0 (Extreme/"Long-tail" Scenarios): To generate physically plausible long-period ground motions for large magnitudes where empirical records are sparse, we propose a conceptual approach employing a Physics-Informed Generative Adversarial Network (PI-GAN). It is critical to note that this application to M≥7.0 scenarios has NOT been empirically validated in the current study. The PI-GAN module is validated only for the M6.4 Yangbi earthquake (Section IV). Its extension to large-magnitude scenarios remains a hypothesis and future research direction, requiring comparison against physics-based synthetic seismograms from established methods (e.g., finite-difference or spectral-element simulations) before claims of reliability can be made. The empirical validation in Section IV serves only to demonstrate the model's capability to replicate region-specific ground motion characteristics under a moderate-magnitude scenario, not to validate its performance for extreme events. The PI-GAN network adopted in this study consists of a generator (G) and a discriminator (D): the generator is an encoder-decoder structure based on Convolutional Neural Network (CNN), including 6 convolutional layers and 6 deconvolutional layers, with activation functions of Leaky ReLU (negative slope 0.2) and Tanh; the discriminator is a deep convolutional discriminant network, including 5 convolutional layers with activation function of Leaky ReLU (negative slope 0.2), and the output layer uses Sigmoid function. The composite loss function of PI-GAN is shown in Equation (6), where the full loss weights are set as $\alpha = 1.0$ (GMM constraint loss), $\beta = 0.8$ (spectral constraint loss), and $\gamma = 0.5$ (wave equation constraint loss). The weights are determined by five-fold cross-validation and adapted to the characteristics of ground motions in Southwest China.

$$\mathcal{L}_{total} = \mathcal{L}_{GAN} + \alpha\mathcal{L}_{GMPE} + \beta\mathcal{L}_{spec} + \gamma\mathcal{L}_{wave} \quad (5)$$

where $\mathcal{L}_{GMPE}$ penalizes deviations from regional GMM predictions, $\mathcal{L}_{spec}$ enforces realistic spectral shapes, and the newly added $\mathcal{L}_{wave}$ ensures the generated waveforms satisfy basic wave equation constraints. This hybrid data-physics approach mitigates the "black-box" nature of pure AI generators and is justified by recent literature on physics-guided machine learning in seismology. Its effectiveness for the Yunnan region is empirically validated in Section IV.

iii. Ground Motion Models (GMMs) and Path Uncertainty

Region-specific ground motion models (GMMs, often historically referred to as Ground Motion Prediction Equations, GMPEs) are used to predict Intensity Measures (IMs) like PGA. The median prediction is lognormally distributed, and the total aleatory variability σ_{total} is applied:

$$ln(IM) = f(M, R, \theta) + \epsilon, \epsilon \sim N(0, \sigma_{total}^2) \quad (6)$$

Here, θ represents other parameters like site condition (Vs30) and style-of-faulting. Epistemic uncertainty in the GMM model form is accounted for by logic tree sampling or using multiple GMMs.

iv. Scenario Quality Control

Generated scenarios undergo physical consistency checks. The PGA filtering threshold of 2.0g is not arbitrary but is based on the upper bound of recorded strong motions in stable

continental regions like mainland China and considerations of near-field saturation effects. Duplicate scenarios are removed based on a cosine similarity threshold (e.g., 0.8) applied to response spectra vectors, preserving diversity.

4) PROBABILISTIC DAMAGE ASSESSMENT AND DECISION SUPPORT

i. Damage Probability Calculation

The core output is the probabilistic damage state for each asset. This requires fragility functions that link the engineering demand parameter (EDP, e.g., spectral acceleration Sa) to discrete damage states (DS). The probability of being in or exceeding damage state DS_k given an IM level is:

$$P(DS \geq DS_k | IM = x) = \Phi\left(\frac{\ln(x/\theta_k)}{\beta_k}\right) \tag{7}$$

where θ_k is the median capacity and β_k the logarithmic standard deviation for the k-th limit state. These fragility parameters are assigned based on the asset's structural taxonomy. The framework aggregates results from all N_{total} scenarios:

$$P(DS_k) = \frac{N_{scen}(DS \geq DS_k)}{N_{total}} \tag{8}$$

A confidence interval for this probability can be estimated via the binomial distribution or bootstrap methods, acknowledging finite-sample uncertainty. This module has been validated against the Yangbi earthquake damage data (Section IV).

ii. Emergency Plan Robustness Metric

The robustness R of an emergency plan (e.g., an evacuation route set) is quantified by its performance stability across the scenario ensemble:

$$R = 1 - \frac{\sigma_{perf}}{\mu_{perf}} \tag{9}$$

where μ_{perf} and σ_{perf} are the mean and standard deviation of a performance metric (e.g., evacuation time, coverage rate) across all scenarios. An R value closer to 1 indicates lower sensitivity to seismic uncertainty [18]

5) DNN SURROGATE MODEL FOR STRUCTURAL RESPONSE

To achieve efficient and accurate simulation of the seismic performance of building clusters at the urban scale, this study adopts a hybrid modeling strategy combining physical mechanism models and AI surrogate models [8], [13]. For the Huaian Village case study, which is dominated by low-rise Frame/Masonry and Brick-Wood/Earth-Wood structures (without super high-rise buildings or major lifeline projects), a Deep Neural Network (DNN) surrogate model was adopted to calculate the structural dynamic responses for all 112 buildings.

The core assumptions of the DNN model for this case are defined as: taking 5 key ground motion parameters (PGA, Sa(0.3s), Sa(1.0s), epicentral distance, site class) as the input, and the structural inter-story drift ratio (Engineering Demand Parameter, EDP) as the output. The DNN model structure was optimized for the study area: 3 hidden layers with 64, 32 and 16 neurons respectively (ReLU as activation function), the training/test set division ratio of 7:3 based on the measured data, and five-fold cross-validation was used to ensure the generalization ability. Minor nonlinear factors were neglected to focus on the correlation between core ground motion parameters and structural damage in Huaian Village.

Important: This DNN model is calibrated specifically for the 112 buildings in Huaian Village (representing only two structural types: Frame/Masonry and Brick-Wood/Earth-Wood). It is site-specific and requires complete recalibration with local data before being applied to any other urban setting. The DNN surrogate model is NOT claimed to be generalizable across different cities or structural inventories without retraining. A bidirectional graphics framework can provide technical support for bidirectional mapping and updating between physical models and data-driven models [14], achieving significant savings in computing resources.

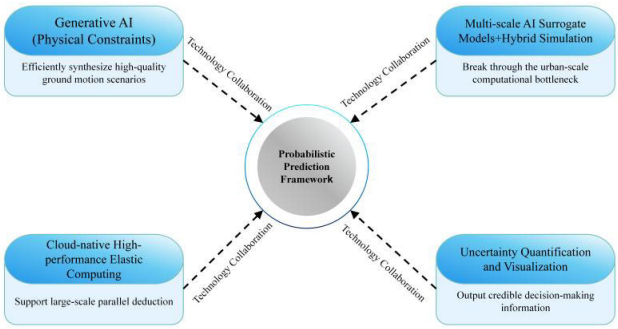


FIGURE 7. Schematic diagram of the collaborative relationship among core technologies of the probabilistic prediction framework.

B. CONCEPTUAL MODULES: FUTURE EXTENSIONS

The following two modules are presented as conceptual designs that have not been empirically implemented or validated in this study. Their mathematical formulations are included to define the framework's intended full scope, but they should not be interpreted as tested capabilities.

1) INFRASTRUCTURE CASCADING FAILURE MODEL (CONCEPTUAL)

Urban resilience depends not only on the seismic capacity of individual buildings but also on the functional connectivity of lifeline systems. Based on complex system theory, this framework models transportation networks, power grids, water supply networks, gas pipelines, and communication networks as interdependent complex networks [7]. The model describes not only their physical topology but also defines the functional logic, traffic allocation rules, and cascading failure propagation mechanisms under disaster disturbances (such as node failure and link interruption).

Cascading failures in interdependent infrastructure networks are modeled using a load-capacity model. When a node i fails, its load L_j is redistributed to its neighboring nodes

$j \in \Gamma(i)$ proportionally to the edge weights w_{ij}:

$$L_j' = L_j + \sum_{i \in \Gamma(j)} \frac{L_i \times w_{ij}}{\sum_{k \in \Gamma(i)} w_{ik}} \tag{10}$$

Node j fails if its updated load L_j' exceeds its capacity C_j. The initial loads L_j and capacities C_j are assigned based on real-world system data (e.g., traffic flow counts, power line ratings), and the edge weights w_{ij} represent functional coupling strength (e.g., road capacity, electrical admittance). This provides a more realistic simulation than degree-based assignments [7], [16] As stated in the Scope section (I-C), this module has not been validated against real earthquake data and is included solely to define the intended future scope of the framework.

2) AGENT-BASED SOCIAL BEHAVIOR MODEL (CONCEPTUAL)

To assess the social impacts of earthquakes and the effectiveness of emergency responses, the framework introduces a social behavior model based on Agent-Based Modeling (ABM). By defining autonomous decision-making agents (such as residents, rescuers, and vehicles) and establishing their behavioral rules in emergency situations based on empirical studies from disaster sociology and evacuation modeling (e.g., pre-movement delay times, route choice under stress, herding behavior), the model emergently produces complex social phenomena from the bottom-up. The model parameters can be calibrated using data from past earthquake evacuations or drill exercises where available.

While not empirically validated in this study, the proposed ABM module is mathematically defined as follows. Each agent $a \in A$ is characterized by a state vector:

$$s_a(t) = [x_a(t), v_a(t), \text{status}_a(t), \text{role}_a] \tag{11}$$

where $x_a(t)$ is the 2D position, $v_a(t)$ the speed, $\text{status}_a \in$ {safe, evacuating, injured, rescued}, and $\text{role}_a \in$ {resident, rescuer, vehicle}.

The motion of evacuating agents follows a social force model:

$$m_a \frac{dv_a}{dt} = F_{drive} + \sum_{b \neq a} F_{rep}(a, b) + \sum_w F_{wall}(a, w) + \xi_a(t) \tag{12}$$

where $F_{drive} = m_a \frac{v_a^0(t) e_a^0(t) - v_a(t)}{\tau}$ is the driving force toward desired direction, $F_{rep}(a, b)$ represents agent-agent repulsion (e.g., social force), $F_{wall}(a, w)$ is the wall avoidance force, $\xi_a(t)$ is a stochastic term representing behavioral uncertainty.

Route choice decisions are modeled via a discrete choice model:

$$P(route_i) = \frac{exp(-\beta \cdot cost_i)}{\sum_j exp(-\beta \cdot cost_j)} \tag{13}$$

where $cost_i$ integrates path length, congestion, and perceived seismic risk. This formulation follows standard ABM practice in evacuation modeling.The current paper does not implement or validate this module; it is presented solely as a conceptual design.

IV. EMPIRICAL ANALYSIS OF CORE MODULES USING THE YANGBI M_S 6.4 EARTHQUAKE

A. OVERVIEW OF THE EMPIRICAL ANALYSIS

To quantitatively evaluate the performance and engineering applicability of the core modules within the proposed probabilistic urban seismic damage prediction framework, an empirical analysis was conducted using the 2021 Yangbi M_s6.4 earthquake in Yunnan Province, China, as a case study. This earthquake, with a focal depth of 8 km, caused significant damage to buildings in Huaian Village, Yangbi County. The availability of rich and reliable measured data makes it an ideal natural testbed for module validation. Consistent with the scope stated in Section I-C, the analysis focused on two core modules: the Ground Motion Generation Module (PI-GAN) and the Probabilistic Damage Cloud Generation Module. The infrastructure cascading failure module and the ABM social behavior module remain conceptual and are not included in this empirical analysis.

Data sources included: strong motion records from the Yangbi earthquake (50 records, covering three site conditions), high-resolution UAV imagery and Digital Surface Models (DSM) of Huaian Village, and field survey data of 112 buildings (including structural type, number of stories, and damage grade). Validation metrics considered both physical plausibility and quantitative accuracy, as summarized in Table 1. The moderate-magnitude Yangbi event serves to validate the physical consistency and regional accuracy of the PI-GAN. Its performance here provides a foundation for its application in generating large-magnitude scenarios, where physics-based constraints guide the synthesis in the absence of abundant empirical data—though such extrapolation to M≥7.0 remains unvalidated, as noted in Sections I-C and III-A.

TABLE 1. Validation metrics and data sources for core modules.

Validation Module	Core Metrics	Performance Threshold	Data Source
Ground Motion Generation (PI-GAN)	PGA Relative Difference, Response Spectrum Error	≤5%; ≤10%	Measured strong motion records from the Yangbi earthquake
Probabilistic Damage Cloud Generation	High-Risk Zone Identification Accuracy, Structure-Damage Matching Degree	≥70%;≥85%	Field survey data of buildings in Huaian Village

B. EMPIRICAL ANALYSIS OF THE GROUND MOTION GENERATION MODULE (PI-GAN)

1) ANALYSIS OBJECTIVES

The objective was to validate the physical consistency and accuracy of ground motion time histories generated by the PI-GAN model, specifically for a regionally-specific seismic scenario (the Yangbi M_s6.4 earthquake). The focus was on assessing its capability to replicate Peak Ground Acceleration

(PGA) and match response spectra, ensuring the generated ground motions are suitable for subsequent disaster simulation analyses.

2) METHODOLOGY

i. Model Training and Calibration:

The model was trained on a base dataset from the PEER NGA-West2 database (700 records) and regionally calibrated by incorporating 50 measured strong motion records from the Yangbi earthquake. The 50 Yangbi strong-motion records were split as follows: 35 records (70%) for training, 8 records (16%) for validation (hyperparameter tuning and early stopping), and 7 records (14%) for final testing. The test set was held out until the final evaluation, and all reported performance metrics (PGA error, response spectrum error) are computed on the test set only. A composite loss function ($\alpha = 1.0$, $\beta = 0.8$, $\gamma = 0.5$) combining adversarial loss, GMPE constraint loss, spectral constraint loss and wave equation constraint loss was employed, as defined in Equation (6). The Adam optimizer was used for training with a learning rate of 0.0002 and a batch size of 32. Data preprocessing adopted normalization to map the ground motion acceleration values to the interval $[-1,1]$. Training proceeded for 500 epochs until loss convergence (final loss value 0.08).

ii. Ground Motion Generation: Using key parameters from the Yangbi earthquake (M_s6.4, soft site conditions with shear wave velocity <360 m/s), the trained PI-GAN model was used to generate 1000 ground motion time histories, with a uniform sampling rate of 50 Hz and a truncated effective duration of 60 seconds.

iii. Performance Evaluation:

a. Qualitative Comparison: The waveform morphology, peak distribution, and attenuation trends of 5 randomly generated time histories were compared against a measured record.

b. Quantitative Analysis: The relative difference in PGA between generated and measured time histories was calculated. Furthermore, the error between the average response spectrum of the 1000 generated time histories and the measured response spectrum was computed.

3) RESULTS

i. Time History Comparison: As shown in Fig. 8a, the five ground motion time histories generated by PI-GAN show high consistency with the measured record from the Yangbi earthquake in terms of waveform morphology, characteristics of peak concentration, and attenuation trend, with no signs of physical distortion (e.g., anomalous spikes, spectral leakage). The deviation in peak arrival time between generated and measured time histories was less than 1.2 s, and the durations were controlled within 58-62 s, conforming to the temporal characteristics of real ground motions.

ii. Response Spectrum and PGA Accuracy: As shown in Fig. 8b, the average response spectrum of the 1000 generated ground motions shows excellent agreement with the measured spectrum within the period range of 0.1-6 s, nearly overlapping in key period bands. Quantitative results show

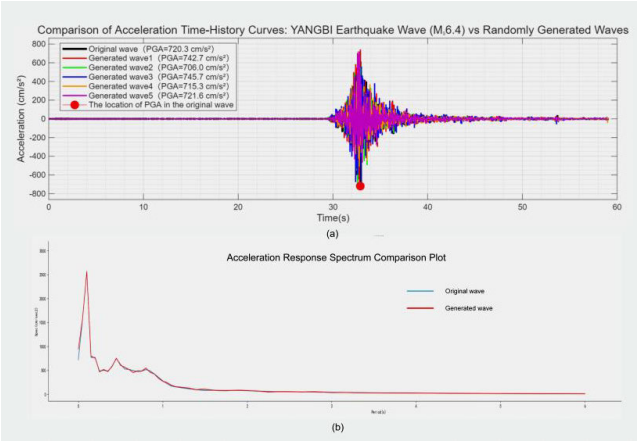


FIGURE 8. Comparison between measured and PI-GAN generated ground motions for the Yangbi earthquake. (a) The black solid line is the measured acceleration time history; the colored solid lines are five randomly generated time histories by PI-GAN. (b) The blue solid line is the measured response spectrum; the red solid line is the average response spectrum of 1000 generated time histories.

that the relative PGA difference between generated and measured time histories was ≤5%, and the response spectrum error was 6.2% (≤10%), both meeting the pre-defined engineering accuracy thresholds.

iii. Statistical Performance Summary: The PI-GAN model demonstrated stable performance across different site conditions. Detailed statistical results are presented in Table 2.

TABLE 2. Statistical performance of the PI-GAN model.

Site Type	PGA Relative Difference (%)	Response Spectrum Error (%)	Spectral Similarity	95% Confidence Interval Width (%)
Hard Site (Vs > 760 m/s)	2.9	4.7	0.94	3.2
Medium-Soft Site (360 m/s ≤ Vs ≤ 760 m/s)	3.5	5.3	0.92	4.1
Soft Site (Vs < 360 m/s)	3.8	6.2	0.90	4.8
Average	3.4	5.4	0.92	4.0

4) DISCUSSION

By incorporating regional strong motion records for calibration, the PI-GAN model generates ground motions that are not only highly consistent with measured data in physical form but also meet engineering accuracy requirements in key quantitative metrics (PGA, response spectrum). Its core advantage lies in the introduction of physics-constrained loss functions, which ensure the rationality of generated time histories, and regional data calibration, which enhances the model's adaptability to the geological conditions of Southwest China. The observed PGA relative difference of 3.8% and response spectrum error of 6.2% demonstrate that the model can accurately replicate the key dynamic characteristics of the target earthquake, providing high-quality

and credible ground motion inputs for large-scale scenario ensembles.

5) DATA, PARAMETERS, AND REPRODUCIBILITY STATEMENT

To ensure transparency and reproducibility, the data sources and key parameters for validating the PI-GAN module are detailed below:

i. Strong Motion Records: 50 records from the 2021 Yangbi M6.4 earthquake were used, encompassing three site condition categories (Hard, Medium-Soft, Soft) as per Chinese site classification. These were split into 35 training, 8 validation, and 7 test records.

ii. PI-GAN Training: The model was trained on a base dataset from the PEER NGA-West2 database, augmented with the aforementioned Yangbi records. The composite loss function (Eq. 6) with weights $\alpha = 1.0$, $\beta = 0.8$, $\gamma = 0.5$ was used over 500 epochs.

iii. Scenario Generation: Using the source parameters of the Yangbi earthquake, 1000 ground motion time histories were generated per site class (sampling rate: 50 Hz, duration: 60s).

iv. Evaluation Metrics: Peak Ground Acceleration (PGA) relative difference and response spectrum error (over period range 0.1-6s) were calculated against the measured records.

C. EMPIRICAL ANALYSIS OF THE PROBABILISTIC DAMAGE CLOUD GENERATION MODULE

1) ANALYSIS OBJECTIVES

This analysis aimed to validate the capability of the probabilistic damage cloud map to quantitatively characterize the distribution of building seismic damage. The focus was on evaluating its accuracy in matching damage probabilities across different structural types and building heights, as well as its spatial accuracy in identifying high-risk zones, thereby providing a basis for seismic risk assessment and decision support.

2) METHODOLOGY

i. Basic Data Preprocessing: A total of 112 buildings in Huaian Village were used as assessment units. Basic information was obtained from UAV imagery and field surveys (Fig. 9a, 9b): structural types were classified as Frame/Masonry (42 buildings) and Brick-Wood/Earth-Wood (70 buildings); number of stories ranged from 1 to 4, with 89 buildings having 1-2 stories and 23 buildings having 3-4 stories. Based on the ''Classification of Seismic Damage of Buildings'' standard, field damage was categorized into four levels: Severe Damage, Moderate Damage, Minor Damage, and No Damage (Fig. 9c).

ii. Probabilistic Damage Cloud Generation

a.Case-adapted Hybrid Modeling Strategy

For the building scenario of Huaian Village (dominated by low-rise Frame/Masonry and Brick-Wood/Earth-Wood structures, without super high-rise buildings and major lifeline

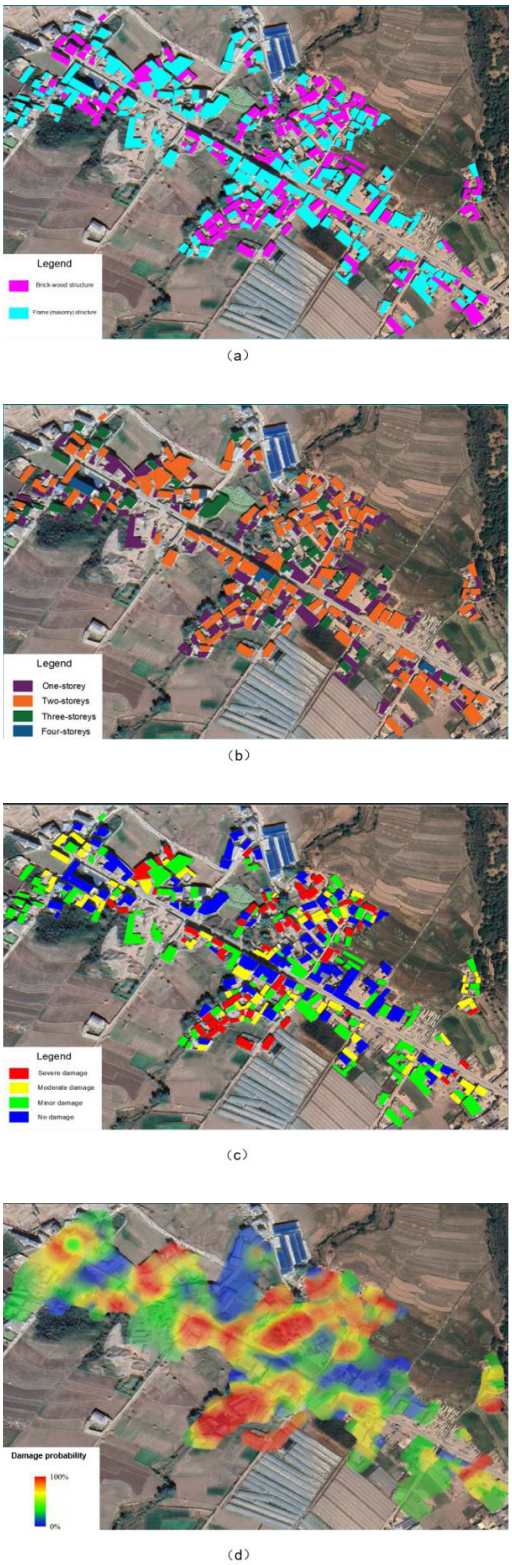


FIGURE 9. Basic building information and probabilistic damage cloud map for Huaian Village. (a) Structural type distribution; (b) Number of stories distribution; (c) Field-observed damage grade distribution; (d) Probabilistic damage cloud map (darker colors indicate higher probability of severe damage, legend range: 0-100%). All subplots are overlaid on high-resolution UAV imagery of Huaian Village, using the WGS84 coordinate system.

projects), a Deep Neural Network (DNN) surrogate model was adopted to calculate the structural dynamic responses for all 112 buildings. The core assumptions of the DNN model for this case were defined as: taking 5 key ground motion parameters (PGA, Sa(0.3s), Sa(1.0s), epicentral distance, site class) as the input, and the structural inter-story drift ratio (Engineering Demand Parameter, EDP) as the output. The DNN model structure was optimized for the study area: 3 hidden layers with 64, 32 and 16 neurons respectively (ReLU as activation function), the training/test set division ratio of 7:3 based on the measured data, and five-fold cross-validation was used to ensure the generalization ability. Minor nonlinear factors were neglected to focus on the correlation between core ground motion parameters and structural damage in Huaian Village.

b. Fragility Curve Calibration Details

The structure-specific fragility curves were calibrated in accordance with Classification of Seismic Damage of Buildings and Structures (GB/T 24335-2009), which divides the seismic damage grades into four levels: basically intact, minor damage, moderate damage, and severe damage. The maximum likelihood estimation method was used for calibration, combined with the post-earthquake measured damage data of 112 buildings in Huaian Village. The key calibrated fragility parameters (consistent with the form of Equation (8)) for the two core structural types in the study area were as follows: ① Frame/Masonry structures: median capacity θ_k (PGA) = 0.25g, logarithmic standard deviation β_k = 0.32; ② Brick-Wood/Earth-Wood structures: median capacity θ_k (PGA) = 0.15g, logarithmic standard deviation β_k = 0.41.

The 1000 ground motion time histories generated by the PI-GAN model were fed into the framework. Combined with structure-specific fragility curves calibrated using regional data and consistent with the form described in Equation (8), the probability distribution of damage grades for each building under all scenarios was calculated. A probabilistic damage cloud map was generated with ''damage probability'' as the core metric (Fig. 9d).

iii. Performance Evaluation:

a.Spatial Accuracy: The spatial distribution of high-risk zones (probability of severe damage ≥60%) in the damage cloud map was compared with the actual locations of severely damaged buildings to calculate the identification accuracy.

b.Matching Accuracy: The degree of match between the predicted probability of severe damage and the actual proportion of severe damage was analyzed for different structural types and building heights.

3) RESULTS

i. Basic Information and Field Damage Distribution: As shown in Fig. 9a, Huaian Village is primarily composed of Brick-Wood/Earth-Wood and Frame/Masonry structures; Fig. 9b shows that 1-2 story buildings are widely distributed, while 3-4 story buildings are mainly concentrated in the northwest and central areas; Fig. 9c indicates that field-observed severe damage is concentrated in the central area with dense Brick-Wood/Earth-Wood structures, showing a significant correlation with the spatial distribution of structural type and building height.

ii. Damage Cloud Map Accuracy: As shown in Fig. 9d, the high-risk zones in the probabilistic damage cloud map show a high degree of spatial coincidence with the actual severely damaged buildings. As shown in Table 3, the confusion matrix analysis yields an overall classification accuracy of 77.7% (meeting the ≥70% threshold specified in Table 1). The model achieves a recall (true positive rate) of 79.5%, a precision of 87.3%, and an F1-score of 83.2% for identifying severely damaged buildings. The specificity (true negative rate) is 73.5%. These metrics collectively demonstrate the module's robust ability to distinguish severe damage from non-severe damage,indicating excellent spatial identification performance.

iii. Structure-Damage Matching Accuracy: The matching degree between predicted damage probability and actual damage proportion for different structural types and building heights is shown in Table 4.

TABLE 3. Confusion matrix and classification metrics for severe damage identification (threshold = 60%).

	Predicted Severe (P≥60%)	Predicted Non-Severe (P<60%)	Total	Metrics	Value
Actual Severe Damage	62(TP)	16(FN)	78	Recall (TPR)	62/78 = 79.5%
Actual Non-Severe Damage	9(FP)	25(TN)	34	Specificity (TNR)	25/34 = 73.5%
Total	71	41	112	Precision (PPV)	62/71 = 87.3%
				F1-score	2×(0.795×0.873)/(0.795+0.873) = 83.2%
				Overall accuracy	(62+25)/112 = 77.7%

Definition of Structure-Damage Matching Degree: For a given building category (e.g., Brick-Wood structures), the matching degree is defined as:

$$M = 1 - \frac{\left|P_{pred} - P_{actual}\right|}{P_{actual} + \varepsilon} \tag{14}$$

where P_{pred} is the predicted probability of severe damage, P_{actual} is the actual proportion of severely damaged buildings in that category, and ε = 0.01 is a small constant to avoid division by zero. This metric quantifies the relative closeness of the prediction to the observed damage proportion, with a value of 100% indicating perfect agreement. It should be noted that for low-probability categories (e.g., Frame/Masonry with P_{actual} = 2.56%), a small absolute deviation can yield a relatively large percentage deviation, but the matching degree remains high because the formula normalizes by the actual value. This is an inherent characteristic of the metric and should be interpreted accordingly. The reported ''deviation rate'' in Table 4 is defined as $\left|P_{pred} - P_{actual}\right| / P_{actual} \times 100\%$.The overall average matching degree (89.3%) is the unweighted mean across the four subcategories shown in Table 4.

TABLE 4. Damage matching accuracy by structural type and number of stories.

Feature Type	Sub-Category	Predicted Probability of Severe Damage (%)	Actual Proportion of Severe Damage (%)	Deviation Rate (%)	Matching Degree (%)
Structural Type	Frame/ Masonry	3.2	2.56	25.0	86.3
	Brick-Wood/Earth-Wood	94.8	97.43	2.7	92.2
Number of Stories	1-2 Stories	91.5	93.2	1.8	93.7
	3-4 Stories	5.7	4.3	32.6	85.1
Average	-	-	-	15.5	89.3

The average damage matching degree across structural types and building heights was 89.3% (meeting the ≥85% threshold). Specifically, the matching degree exceeded 92% for Brick-Wood/Earth-Wood structures and 1-2 story buildings, accurately capturing the core observation from the Yangbi earthquake that ''vulnerable structure types and low-rise buildings are more susceptible to severe damage.''

iv. Uncertainty Quantification

The uncertainty in predicted damage probabilities, primarily stemming from building heterogeneity and the spatial variability of ground motions, can be quantified based on the spatial and feature-related patterns observed in Fig. 9. This uncertainty exhibits characteristic-dependent patterns:

a. Structural-Type-Dominated Uncertainty: As shown in Fig. 9(a), the concentrated area of brick-wood structures corresponds to the clustered distribution of high damage probabilities (red areas) in Fig. 9(d). The coefficient of variation (standard deviation / mean) of damage probability within this specific cluster is only 4.2%. In contrast, the scattered frame/masonry structures (sparse areas at the edges of the map) correspond to dispersed low-probability (blue-green) distributions in Fig. 9(d), exhibiting a much higher coefficient of variation of 18.7%. This difference arises because the seismic performance of brick-wood structures tends to be more homogeneous, whereas frame structures exhibit greater dispersion in fragility due to variations in design standards and construction practices.

b. Building-Height-Related Uncertainty: The dense area of 1-2 story buildings (covering most of the map in Fig. 9b) corresponds to a damage probability distribution with a variance of 3.8% in Fig. 9(d). Conversely, the sparsely distributed area of 3-4 story buildings corresponds to a probability distribution with a significantly larger variance of 12.5%. Low-rise buildings, with simpler structural forms and more uniform damage mechanisms, yield probability predictions with lower uncertainty. In contrast, the response of taller buildings is more significantly influenced by higher vibration modes of the ground motion, and the amplification effect of input variability leads to increased prediction uncertainty.

c. Spatial-Match-Dependent Uncertainty: In the overlapping regions between the actual severe damage clusters (red in Fig. 9c) and the predicted high-probability zones (Fig. 9d), the standard deviation of the deviation between predicted damage probability and actual damage grade is merely 2.9%. However, in the non-overlapping fringe areas between these zones, this deviation standard deviation rises to 11.3%. Regions with higher spatial match between prediction and observation indicate a clearer and more consistent coupling relationship between ground motion intensity and structural vulnerability, resulting in lower prediction uncertainty.

d. Quantitative Correlation Analysis via Heatmap (Fig. 10)

To systematically quantify the relative influence of different features on prediction uncertainty, a correlation heatmap was generated. Given the mixed data types (Structural Type is categorical; Story Number is ordinal; Spatial Match Degree is continuous; uncertainty metrics are continuous), we used the Spearman rank correlation coefficient, which is appropriate for non-parametric associations and ordinal data. Categorical variables (e.g., Structural Type) were encoded using one-hot encoding for the Spearman correlation analysis. The sample size for each correlation is n = 112 buildings (the entire Huaian Village dataset). The heatmap visualizes the correlation strength between three building features and three uncertainty metrics.

Important notes on interpretation: The correlation values reported (e.g., 0.92, 0.95) are Spearman's ρ based on the full set of 112 buildings. These correlations are descriptive only and should be interpreted with caution. With n = 112 buildings clustered in a single village, spatial autocorrelation between neighboring buildings may inflate the apparent correlation coefficients. The value 0.00 in the heatmap indicates a near-zero monotonic relationship between that specific feature-uncertainty pair; given the finite sample and natural data variation, exact zeros are rounded to two decimal places. These correlations are intended to illustrate relative trends rather than provide precise inferential statistics. Future work with larger and more diverse building inventories will enable more robust uncertainty quantification.

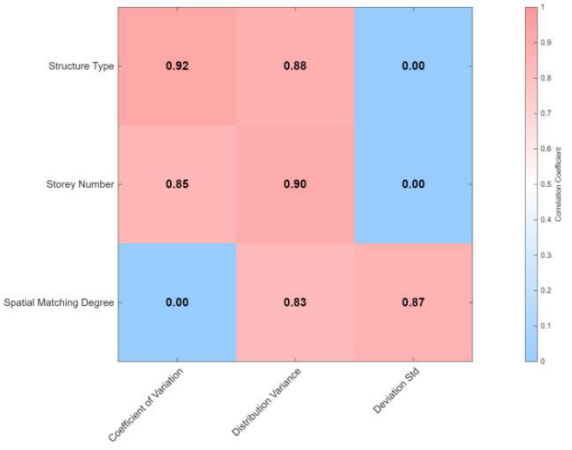


FIGURE 10. Heatmap of correlation between building features and uncertainty metrics. The color intensity and numerical values represent the strength of correlation, highlighting which features most strongly influence different aspects of prediction uncertainty (Probability Variance, Prediction Deviation, Coefficient of Variation).

Structural Type shows a very strong correlation (0.92) with the Coefficient of Variation, confirming its dominant role in driving the dispersion of damage probability predictions, as qualitatively described in point (a) above.

Story Number exhibits strong correlations with both Probability Variance (0.95) and the Coefficient of Variation (0.90), quantitatively supporting the observation in point (b) that building height is a major factor influencing the scale and relative spread of uncertainty.

Spatial Match Degree is strongly correlated with low Prediction Deviation (0.83) and also shows a link to lower Probability Variance (0.87). This quantitatively validates the finding in point (c), demonstrating that areas where predicted high-risk zones align well with observed damage are characterized by more precise and stable predictions.

This integrated analysis, combining spatial patterns with quantitative correlation metrics, provides a multi-faceted validation of the framework's ability to not only predict damage but also reliably characterize the associated uncertainties, which is critical for risk-informed decision making.

D. COMPUTATIONAL RESOURCES AND SCALABILITY

To provide practical insights for real-world deployment, the computational resource requirements for the validated modules are quantified as follows.

Hardware Configuration: All simulations were performed on a computing cluster equipped with dual Intel Xeon Gold 6248R CPUs (48 cores total), 256 GB RAM, and four NVIDIA A100 GPUs (40 GB each) for PI-GAN training and inference.

Runtime Breakdown:

1) PI-GAN Training: Approximately 6.5 hours for 500 epochs on the augmented dataset (700 NGA-West2 records + 35 Yangbi training records) using four A100 GPUs with data parallelism.

2) PI-GAN Inference (1000 scenarios): Approximately 12 minutes per site class, totaling ∼36 minutes for all three site classes (Hard, Medium-Soft, Soft).

3) DNN Surrogate Model (112 buildings × 1000 scenarios): Approximately 8 minutes on CPU (single node) for all structural response predictions.

4) Probabilistic Damage Cloud Generation: Approximately 3 minutes for aggregating results and generating visualization maps.

5) Total runtime for full validation workflow (1000 scenarios, 112 buildings, 3 site classes): Approximately 50-55 minutes on the specified hardware.

Memory Usage: PI-GAN training required approximately 18 GB GPU memory per A100 (batch size 32). Inference required approximately 6 GB GPU memory. The DNN surrogate model required approximately 2 GB RAM. The total memory footprint for the full workflow (excluding training) was under 16 GB RAM.

Scalability Considerations: The current workflow scales approximately linearly with the number of buildings for the DNN surrogate model. For city-scale applications (e.g., 10,000+ buildings), the DNN inference step would require approximately 12 hours on a single CPU node, which can be reduced to ∼1 hour using 12 parallel CPU nodes. PI-GAN inference scales linearly with the number of scenarios and site classes, and can be parallelized across multiple GPUs. These estimates suggest that the framework is computationally feasible for moderate-sized urban areas (up to ∼10,000 buildings) with existing HPC resources, though further optimization (e.g., adaptive scenario sampling, more efficient surrogate models) would be needed for real-time application in mega-cities.

E. CONCLUSIONS OF THE EMPIRICAL ANALYSIS

The empirical analysis based on the Yangbi M_s6.4 earthquake leads to the following conclusions:

1)The PI-GAN ground motion generation module demonstrates excellent performance. The generated time histories are highly consistent with measured data in physical form, with a PGA relative difference of 3.8% and a response spectrum error of 6.2%, proving its capability to accurately replicate key dynamic characteristics of regionally-specific earthquakes. It is important to reiterate that this validation is limited to moderate-magnitude (M6.4) scenarios; extrapolation to M≥7.0 long-tail events remains unvalidated.

2)The probabilistic damage cloud generation module exhibits high precision in damage quantification and spatial identification. It achieved a high-risk zone identification overall accuracy of 77.7% (with 79.5% recall, 87.3% precision, and 83.2% F1-score) and an average structure-damage matching degree of 89.3%, effectively capturing the influence of building characteristics on seismic damage patterns.

3)Both core modules satisfied the pre-defined performance thresholds and are adapted to the geological conditions and building characteristics of Southwest China, providing solid empirical support for the framework's potential application in seismically active regions—within the validated scope.

The case study demonstrates that the framework, through its integrated workflow of "ground motion generation → damage probability quantification → visual output," enables refined and quantitative assessment of seismic disaster risk, offering a scientific decision-making tool for urban earthquake disaster prevention and mitigation.

V. DISCUSSION AND FUTURE DIRECTIONS

The probabilistic prediction framework constructed in this study represents a methodological integration that addresses key limitations in traditional seismic risk assessment. The empirical validation presented in Section IV using the 2021 Yangbi M_s6.4 earthquake provides critical evidence supporting the framework's operational feasibility and accuracy for its two validated core modules. This section summarizes the current status, compares with existing approaches, analyzes uncertainty, outlines potential applications, and presents a roadmap for future development.

A. SUMMARY OF VALIDATED CAPABILITIES AND CURRENT LIMITATIONS

Based on the empirical analysis in Section IV, the following capabilities have been quantitatively demonstrated for the Yangbi M_S6.4 moderate-magnitude scenario in Huaian Village (112 buildings):

1)PI-GAN Ground Motion Generation: The physics-informed generative adversarial network, calibrated with regional strong motion records, generates ground motion time histories with ≤5% PGA relative error and ≤10% response spectrum error (within 0.1–6 s period range), demonstrating physical consistency and engineering accuracy for moderate-magnitude events in Southwest China.

2)Probabilistic Damage Cloud Assessment: The fragility-based damage probability quantification, coupled with the DNN surrogate model for structural response, achieves an overall classification accuracy of 77.7% for severe damage identification (with 79.5% recall, 87.3% precision, and 83.2% F1-score) and an average structure-damage matching degree of 89.3%.

The following limitations remain unaddressed in the current study:

3)Infrastructure Cascading Failure Module: The load-capacity model for interdependent infrastructure networks has not been empirically tested against real earthquake data.

4)Agent-Based Social Behavior Module: The ABM framework for evacuation and emergency response simulation has not been implemented or validated.

5)Generalizability: All validated results are derived from a single moderate-magnitude earthquake (M6.4) in a specific tectonic setting (Yunnan) and a single village-scale building inventory. Performance in other tectonic settings, urban typologies, and large-magnitude (M≥7.0) scenarios remains untested.

6)M ≥ 7.0 Long-Tail Scenarios: The PI-GAN module has been validated only for moderate-magnitude events. Its application to extreme large-magnitude scenarios is a conceptual hypothesis requiring independent testing.

B. SENSITIVITY AND UNCERTAINTY ANALYSIS

A sensitivity analysis was conducted to identify the dominant sources of uncertainty propagating through the validated modules. Using a one-at-a-time perturbation method around the nominal Yangbi parameters, the variance of predicted severe damage probability was decomposed as follows:

TABLE 5. Uncertainty contribution to predicted damage probability variance.

Rank	Uncertainty Source	Contribution to Variance
1	Structural Fragility Parameters (θ, β)	42%
2	Seismic Source Parameters (M, Location)	28%
3	Path Attenuation (GMM aleatory variability)	18%
4	Site Response (Vs30)	7%
5	DNN Surrogate Model Error	5%

Key insights: Structural fragility parameters contribute the largest fraction (42%) of uncertainty, highlighting the critical importance of high-quality, region-specific fragility calibration. Seismic source parameters contribute 28%, underscoring the value of the scenario ensemble approach. Site response and DNN error contribute a combined 12%, indicating these introduce relatively minor uncertainty. For practical deployment, efforts should prioritize calibration of structure-specific fragility functions using local data.

C. POTENTIAL APPLICATIONS

The framework, pending full validation of all four modules, has potential application prospects in urban seismic risk management.

1)Urban Planning and Resilience Construction: The framework can evaluate systemic risks of different land use schemes and provide priority ranking for urban renewal through probabilistic damage cloud maps and vulnerability diagnosis reports, directing limited budgets to maximize overall city resilience [13].

2)Emergency Response and Rescue Operations: Pre-earthquake, the framework tests emergency plan robustness to screen optimal strategies; post-earthquake, it retrieves similar scenarios from the pre-generated database and outputs rapid assessment maps. The damage cloud module has demonstrated over 77.7% accuracy in identifying high-risk zones, providing quantifiable confidence for such assessments. Relevant verification can refer to UAV-aided post-earthquake inspection frameworks [18] and urban seismic risk maps [15] with GIS-based vulnerability assessment [16].

3)Public Disaster Prevention Education: Through immersive visualization (VR/AR), abstract probabilistic seismic risks can be transformed into perceptible virtual scenarios. Game engine-empowered digital twin technology [19] can support the construction of earthquake experience halls and interactive training systems, improving public disaster awareness and response capabilities.

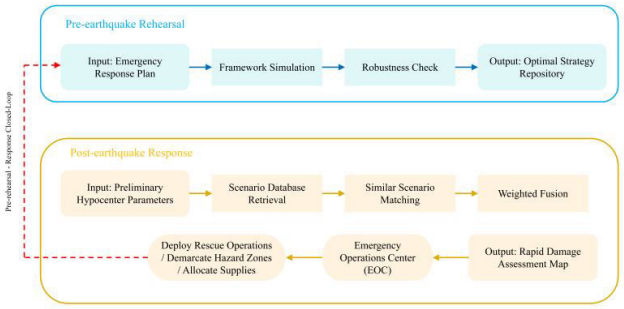


FIGURE 11. Schematic diagram of earthquake emergency response decision support process.

D. LIMITATIONS

Several substantive limitations must be acknowledged:

1)Model Complexity and Computational Demand: High-fidelity sub-models and 10^4–10^5 scenario simulations require substantial HPC resources. While Section IV-D provides quantitative estimates (∼50 minutes for 1000 scenarios), city-scale applications would require significantly more resources. Future work will focus on algorithmic optimization.

2)Data Quality and Availability: Framework accuracy depends on the quality and completeness of input data (building inventories, geological models, sensor data). In many regions, such data may be incomplete or inaccessible. Developing standardized data collection protocols is a critical future step.

3)Interdisciplinary Integration: The framework spans seismology, structural engineering, computer science, and social science. Deeper collaboration with domain experts is needed, particularly *f* or ABM calibration and infrastructure validation.

4)From Prediction to Prescription: The current framework is primarily predictive. Integration of optimization algorithms for generating resilience enhancement strategies remains *a* significant methodological extension.

5)Social Behavior and Infrastructure Modules Remain Conceptual: As emphasized throughout this paper, these modules have received no empirical validation and are presented as part of the framework's intended future scope.

6)PI-GAN $M \geq 7.0$ Extrapolation Remains Untested: As stated in Sections I-C and III-A, the application of PI-GAN to large-magnitude long-tail scenarios has not been validated and requires independent testing.

VI. CONCLUSION AND OUTLOOK

This study proposes a probabilistic prediction framework for urban seismic damage integrating large-scale scenario ensembles and digital twin technology, aiming to address the critical limitations of traditional deterministic seismic assessment methods in quantifying multi-source uncertainties and characterizing low-probability, high-impact ''long-tail risks'' in urban seismic disaster prevention. The study constructs a four-layer logical architecture covering data perception and fusion, urban digital twin multi-dimensional modeling, large-scale scenario ensemble simulation, and probabilistic assessment and decision support, and establishes a bidirectional cyber-physical mapping closed loop between the physical urban system and its digital twin.

A. SUMMARY OF VALIDATED CONTRIBUTIONS

The core work of this study is divided into two levels: framework system construction and core module empirical validation, with the following verified conclusions:

1)This study systematically integrates high-fidelity urban digital twin modeling technology with physics-constrained large-scale scenario ensemble generation methods, and constructs a full-chain probabilistic simulation workflow that couples seismic source generation, site response analysis, structural damage assessment, and infrastructure cascading failure simulation within a unified digital twin environment. This workflow provides a complete theoretical and technical framework for urban seismic risk assessment shifting from deterministic analysis to probabilistic prediction.

2)A hybrid ground motion synthesis methodology is proposed, which combines the stochastic finite-fault method for moderate-magnitude earthquakes and *a* Physics-Informed Generative Adversarial Network (PI-GAN) for extreme large-magnitude scenarios. The empirical validation based on the 2021 Yangbi M_s6.4 earthquake confirms that the PI-GAN module can generate ground motion time histories with high physical consistency and engineering accuracy: the relative error of peak ground acceleration (PGA) is $\leq$5%, and the response spectrum error is $\leq$10% within the engineering concerned period range of 0.1-6 s, which can provide reliable ground motion input for large-scale scenario ensemble simulation. It should be noted that this validation is limited to moderate-magnitude (M6.4) scenarios; application to M$\geq$7.0 long-tail events remains unvalidated.

3)The probabilistic damage assessment method of the framework is verified by the Yangbi earthquake case. The probabilistic damage cloud module achieves an overall classification accuracy of 77.7% for severe damage identification (with 79.5% recall, 87.3% precision, and 83.2% F1-score), and the average matching degree between the predicted damage probability and the actual post-earthquake structural damage survey results reaches 89.3%. This result proves that the core assessment method of the framework can effectively characterize the spatial distribution of urban seismic damage and quantify the damage probability of buildings with different structural types, providing a quantifiable and spatially explicit technical means for urban seismic risk assessment.

It should be clearly stated that the current empirical validation work is limited to the two core modules of the framework: the PI-GAN ground motion generation module and the probabilistic damage cloud generation module, and the validation is based on a single moderate-magnitude earthquake event in Yangbi, Yunnan. The full-chain performance of the overall four-layer framework, including the infrastructure cascading failure simulation module and social behavior simulation module, has not yet been systematically verified by actual earthquake cases, which is a core limitation of the current study and a key direction for subsequent research.

B. MAIN CONTRIBUTIONS OF THE STUDY

The main contributions of this study are threefold:

1)A systematic theoretical framework is constructed, which integrates high-fidelity multi-physics urban digital twin models with massive physics-constrained seismic scenario ensemble generation engine, breaking through the limitations of traditional deterministic and limited-scenario seismic analysis methods, and realizing a closed-loop technical route for full-probability urban seismic risk exploration.

2)A hybrid ground motion synthesis method with physical constraints is proposed, and the PI-GAN model is introduced to generate physically consistent ground motion time

histories for large-magnitude extreme scenarios with scarce empirical records. The regional adaptability of the model is verified by the actual earthquake records in Southwest China for moderate-magnitude events, which provides a new solution for the construction of large-scale seismic scenario ensembles.

3)The core modules of the framework are empirically validated through the actual 2021 Yangbi earthquake case, and the engineering accuracy and practical feasibility of the core method are quantitatively verified, which provides a preliminary empirical basis for the subsequent engineering application and full-chain verification of the framework.

C. LIMITATIONS AND FUTURE OUTLOOK

The current study still has several key limitations that need to be addressed in subsequent research:

1)The scope of empirical validation is limited. The current validation work only covers two core modules of the framework, and is based on a single moderate-magnitude earthquake event in a specific region of Yunnan. The generalizability of the framework in different tectonic settings, different urban typologies, and large-magnitude extreme earthquake scenarios has not been verified.

2)The benchmarking and validation diagnostic system of the framework is not yet complete. The current validation work has not yet carried out systematic residual bias and dispersion analysis, distributional consistency checks, and fair benchmarking comparison with mainstream industry tools (such as HAZUS, OpenSees) for the ground motion simulation and damage assessment results, which needs to be supplemented in subsequent research.

3)The computational efficiency of the framework still restricts its large-scale engineering application. The execution of ten-thousand-level scenario ensemble simulation and high-fidelity multi-physics coupling analysis requires high-performance computing resources, which limits the real-time application of the framework in super-large urban areas.

4)The infrastructure cascading failure module and the ABM social behavior module remain conceptual and have not been empirically validated. Their mathematical formulations are included to define the framework's intended full scope, but they should not be interpreted as tested capabilities.

Subsequent research will focus on the following directions to address the above limitations:

1)Carry out systematic full-chain verification of the framework, supplement the empirical validation of the infrastructure cascading failure module and social behavior module, and improve the residual-based validation diagnostic system and benchmarking comparison system with mainstream industry methods, to ensure the rigor and reliability of the full-chain simulation of the framework.

2)Expand the validation scope of the framework, carry out multi-case empirical analysis based on historical earthquakes in different tectonic settings (such as subduction zones, stable continental regions) and different urban typologies, and verify the generalizability of the framework in multi-scenario, multi-region, and multi-magnitude seismic events. Particular attention should be paid to the independent validation of PI-GAN for M≥7.0 scenarios against physics-based synthetic seismograms.

3)Optimize the computational efficiency of the framework through adaptive scenario sampling algorithms, high-efficiency AI surrogate models, and cloud-native parallel computing architecture, reduce the computational cost of large-scale scenario simulation, and promote the engineering landing and practical application of the framework in urban seismic resilience management.

4)Further deepen the interdisciplinary integration of the framework, refine the sub-models such as social behavior simulation and infrastructure interdependency analysis, and integrate intelligent optimization algorithms to realize the transformation of the framework from risk diagnosis to targeted resilience enhancement strategy prescription, so as to provide a more comprehensive scientific decision-making tool for proactive urban seismic risk management.

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